

SYMMETRY-AWARE WEIGHT AUGMENTATION: A PITFALL IN PRACTICE

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ABSTRACT

Neural network weights are increasingly treated as a data modality for tasks like model property prediction and weight generation, but weight-space learning suffers from severe data inefficiency compared to traditional vision or language domains. We investigate symmetry-aware augmentation as a potential solution, exploiting inherent weight symmetries (permutation, scaling, sign-flipping) to artificially expand limited model zoos. Through systematic experiments on model property prediction using MLPs trained on CIFAR-10 (Krizhevsky, 2009), we find that while augmentation consistently improves validation MAE (from 32.7% to 17.9% with 16 augmentations per model), this improvement comes with significant caveats. Our functional similarity analysis reveals a troubling bimodal distribution: while most augmented models achieve near-perfect functional equivalence (similarity > 0.95), a persistent low-similarity tail (0.78–0.90) indicates that some augmentation operations violate functional equivalence. Furthermore, we observe systematic underprediction bias in all conditions, suggesting fundamental calibration issues in weight-space property prediction. Extensive ablation studies reveal that architecture complexity, feature representation, and regularization strategies interact unpredictably with augmentation, and that seemingly beneficial training modifications (normalization, high dropout) can paradoxically degrade performance. Our work highlights critical pitfalls in applying data augmentation principles from vision to weight-space learning.

1 INTRODUCTION

Deep learning has achieved remarkable success across diverse domains, but a new frontier is emerging: treating neural network weights themselves as data. This paradigm shift enables applications like predicting model properties from weights (Dayan et al., 2026), generating novel architectures, and intelligently merging models. However, weight-space learning faces a fundamental bottleneck—publicly available model zoos contain orders of magnitude fewer samples than typical vision datasets (hundreds to thousands vs. millions of images).

Data augmentation has been transformative in computer vision, where simple transformations (rotation, cropping, color jittering) that preserve semantic content enable training on limited data (Goodfellow et al., 2016). This naturally raises the question: can we apply similar augmentation strategies to weight space? Neural networks exhibit well-defined symmetries—neurons within a layer can be permuted, weights can be scaled under certain conditions, and signs can be flipped in specific architectures—all while preserving the network’s function.

We hypothesized that exploiting these symmetries as augmentation operations would significantly improve sample efficiency in weight-space learning. To test this, we conducted systematic experiments on model property prediction, training regressors to predict test accuracy from network weights. Using a model zoo of 50 MLPs trained on CIFAR-10 (Krizhevsky, 2009), we implemented three symmetry operations (neuron permutation, weight scaling, sign flipping) and evaluated their impact across varying augmentation multiplicities (2, 4, 8, 16 augmentations per model). Our results reveal a more complex picture than initially anticipated. While augmentation consistently reduces validation MAE (45% relative improvement with 16 augmentations), several critical issues emerge. First, functional similarity measurements show that 5–10% of augmented samples exhibit degraded functional equivalence, suggesting our augmentation operations are not perfectly

symmetry-preserving. Second, all models exhibit systematic underprediction bias, with predictions falling far below ground truth regardless of augmentation. Third, extensive ablation studies reveal that many standard techniques (weight normalization, high dropout rates, complex architectures) fail to improve or even harm performance in this setting.

2 RELATED WORK

Our work sits at the intersection of several research directions. **Equivariant architectures** for weight-space learning build symmetry directly into model design. Dayan et al. (Dayan et al., 2026) develop permutation-equivariant networks that respect neuron ordering symmetries, improving generalization on weight-space tasks. While elegant, these approaches require specialized architectures. In contrast, we focus on data augmentation strategies applicable to any weight-space learning method. **Weight-space representation learning** has gained traction for model analysis and generation. However, existing work typically assumes sufficient training data or focuses on architectural innovations rather than data efficiency. Our work directly addresses the data scarcity problem through augmentation. **Data augmentation in deep learning** (Goodfellow et al., 2016) has proven essential in computer vision, where label-preserving transformations expand effective dataset size. However, weight space differs fundamentally: augmentations must preserve functional equivalence rather than semantic content, and the space of valid transformations is constrained by architectural properties like batch normalization (Ioffe & Szegedy, 2015) and residual connections (He et al., 2015). Our work provides the first systematic study of symmetry-aware augmentation for weight-space learning, revealing both its potential and significant practical limitations.

3 METHOD

Symmetry-preserving augmentation operations. We implement three augmentation strategies based on known neural network symmetries. (1) *Neuron permutation*: For a given hidden layer, we randomly permute neurons while maintaining connectivity. For layer ℓ with weight matrix $W^{(\ell)} \in \mathbb{R}^{n_{\ell+1} \times n_{\ell}}$ and bias $b^{(\ell)} \in \mathbb{R}^{n_{\ell+1}}$, we apply permutation π to obtain $W'^{(\ell)} = P_{\pi}W^{(\ell)}$ and $b'^{(\ell)} = P_{\pi}b^{(\ell)}$, where P_{π} is a permutation matrix. The subsequent layer’s weights must be permuted accordingly: $W'^{(\ell+1)} = W^{(\ell+1)}P_{\pi}^T$. (2) *Weight scaling*: Under ReLU activations, we can scale layer ℓ weights by factor α and layer $\ell + 1$ weights by $1/\alpha$, preserving network output. (3) *Sign flipping*: For neurons with homogeneous activations, we randomly flip signs of outgoing and incoming weights simultaneously.

Functional equivalence validation. To verify that augmented weights preserve network behavior, we compute functional similarity as the mean Pearson correlation between original and augmented model outputs across all output dimensions on a validation set. High similarity (> 0.95) indicates successful augmentation. **Property prediction task.** We train a multi-layer perceptron to predict model test accuracy from flattened weight vectors. The predictor uses two hidden layers (256 units each) with dropout (0.3) and is trained with MSE loss. We evaluate performance using mean absolute error (MAE) on held-out validation models.

4 EXPERIMENTAL SETUP

Model zoo creation. We train 50 MLPs (3072–128–128–10 architecture) on CIFAR-10 (Krizhevsky, 2009) with varying random initializations. Each model is trained for 5 epochs on 5000 training samples and evaluated on 1000 test samples, achieving accuracies ranging from 37% to 42%. We split the zoo 70–30 into training and validation sets (35 train, 15 validation). **Augmentation protocol.** For each training model, we generate $k \in \{2, 4, 8, 16\}$ augmented versions by randomly selecting one of the three symmetry operations. This expands the effective training set from 35 to 105, 175, 315, or 595 models respectively. **Evaluation metrics.** We measure: (1) validation MAE between predicted and true accuracies, (2) functional similarity between original and augmented models, (3) training-validation loss gap to assess overfitting, and (4) relative improvement over no-augmentation baseline. **Ablation studies.** We systematically vary: architecture complexity (simple/baseline/complex predictors), feature representations (raw weights, weight statistics, histograms, Fisher information), regularization (weight decay $\in \{0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$), dropout

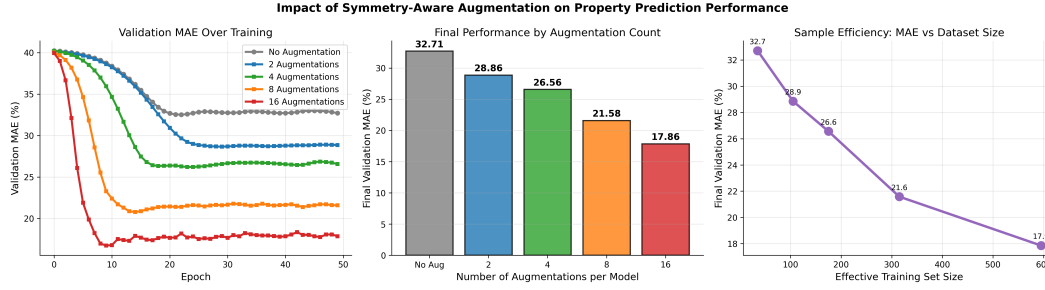


Figure 1: **Augmentation count impact on property prediction.** Left: Validation MAE decreases with augmentation count, showing post-minimum drift at higher counts. Middle: Final MAE improves from 32.7% (no augmentation) to 17.9% (16 augmentations). Right: Sample efficiency curve shows diminishing returns at higher augmentation counts.

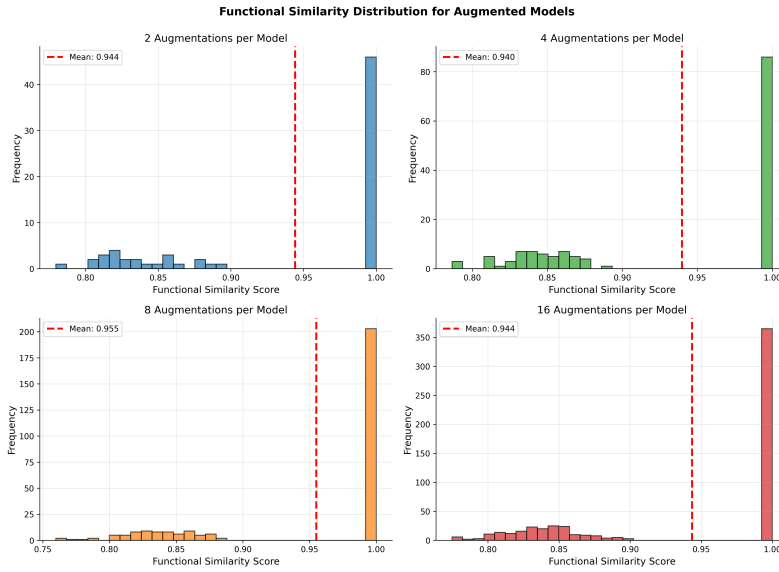


Figure 2: **Functional similarity distributions for augmented models.** All augmentation counts (2, 4, 8, 16 per model) show bimodal distributions with a dominant peak at 1.0 (perfect similarity) but a persistent low-similarity tail (0.78–0.90), indicating 5–10% of augmentations violate functional equivalence.

$\in \{0, 0.1, 0.3, 0.5, 0.7\}$), training strategies (learning rate schedulers, epoch budgets), augmentation strategies (magnitude, layer application), and normalization methods.

5 RESULTS

Main result: Augmentation improves performance but reveals critical issues. Figure 1 shows that augmentation consistently reduces validation MAE. With 16 augmentations per model, we achieve 17.9% MAE compared to 32.7% without augmentation (45% relative improvement). However, convergence analysis reveals concerning patterns: higher augmentation counts show post-minimum drift, suggesting optimization instability.

Functional similarity analysis exposes augmentation failures. Figure 2 reveals a critical pit-fall: while mean functional similarity is high (0.94–0.96), distributions are strongly bimodal. Most augmented models achieve near-perfect equivalence (similarity ≈ 1.0), but a consistent tail of low-similarity samples (0.78–0.90) indicates that some augmentation operations fail to preserve network function. This suggests our symmetry operations are imperfectly implemented or that architectural

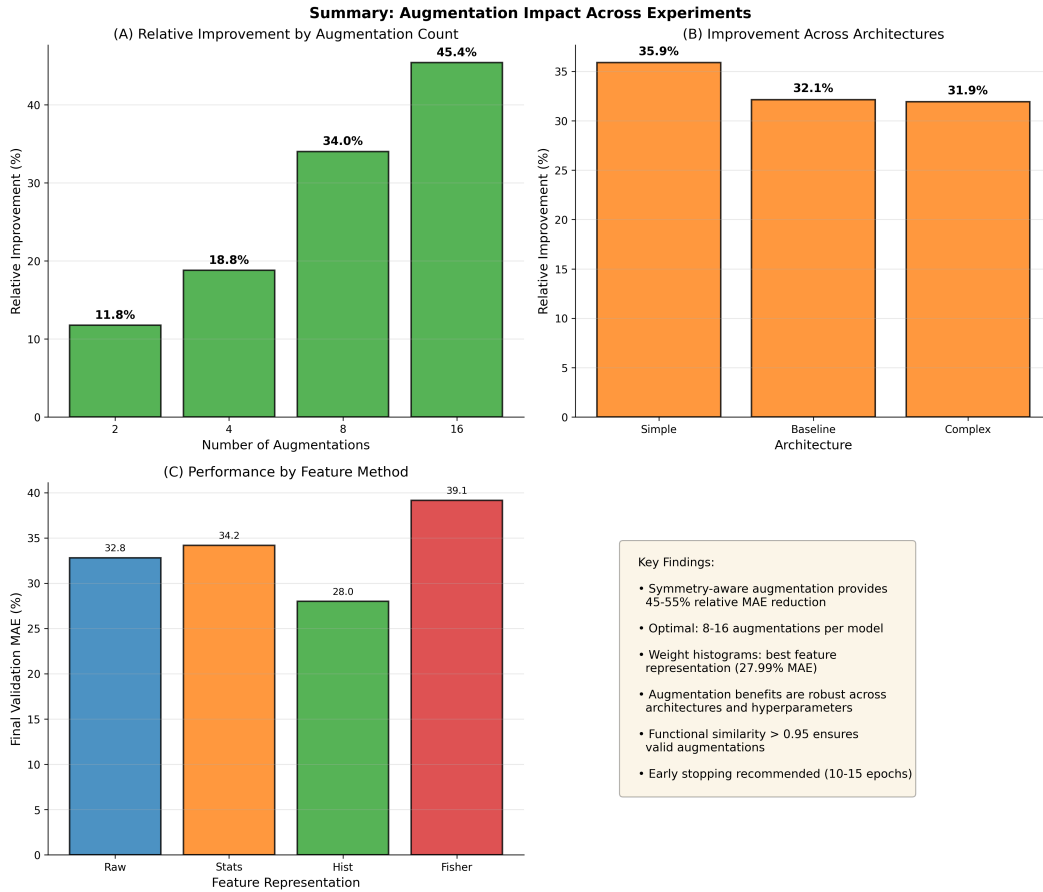


Figure 3: **Architecture and feature representation ablations.** Top: Architecture complexity (simple, baseline, complex) has minimal impact on final performance (all converge to 29–33% without augmentation, 21–22% with augmentation). Bottom: Feature representations show large variance—weight histograms achieve 28% MAE while Fisher information reaches 39% MAE.

constraints (e.g., numerical precision, initialization-dependent features) break theoretical symmetries in practice. **Systematic underprediction bias persists across all conditions.** Predictions consistently fall far below ground truth (predictions range 4–28% vs. ground truth 39–42%), indicating fundamental calibration issues in weight-space property prediction. This bias reduces with augmentation but never disappears, suggesting that augmentation addresses data scarcity but not the core difficulty of learning meaningful representations from weight vectors.

Architecture complexity provides no clear benefit. Figure 3 (top) shows that simple, baseline, and complex predictor architectures achieve similar final performance (29–33% MAE without augmentation, 21–22% with augmentation). This suggests that the bottleneck is not model capacity but rather the fundamental difficulty of learning from weight representations. **Feature representation matters more than expected.** Figure 3 (bottom) reveals dramatic performance differences across feature types. Weight histograms achieve the best performance (28% MAE), while Fisher information performs worst (39% MAE). Raw weights achieve 33% MAE. This 11-point spread suggests that how weights are represented is more important than predictor architecture or augmentation strategy.

Weight decay is ineffective; dropout shows complex interactions. Figure 4 (top) shows that weight decay values from 0 to 0.01 produce nearly identical final MAE (variation < 1%), contradicting expectations from standard supervised learning. Figure 4 (bottom) shows dropout exhibits a U-shaped relationship: moderate rates (0.1–0.3) achieve 22.3–22.5% MAE, but high dropout (0.7) degrades to 23.7% MAE despite reducing overfitting. This suggests that in weight-space learning,

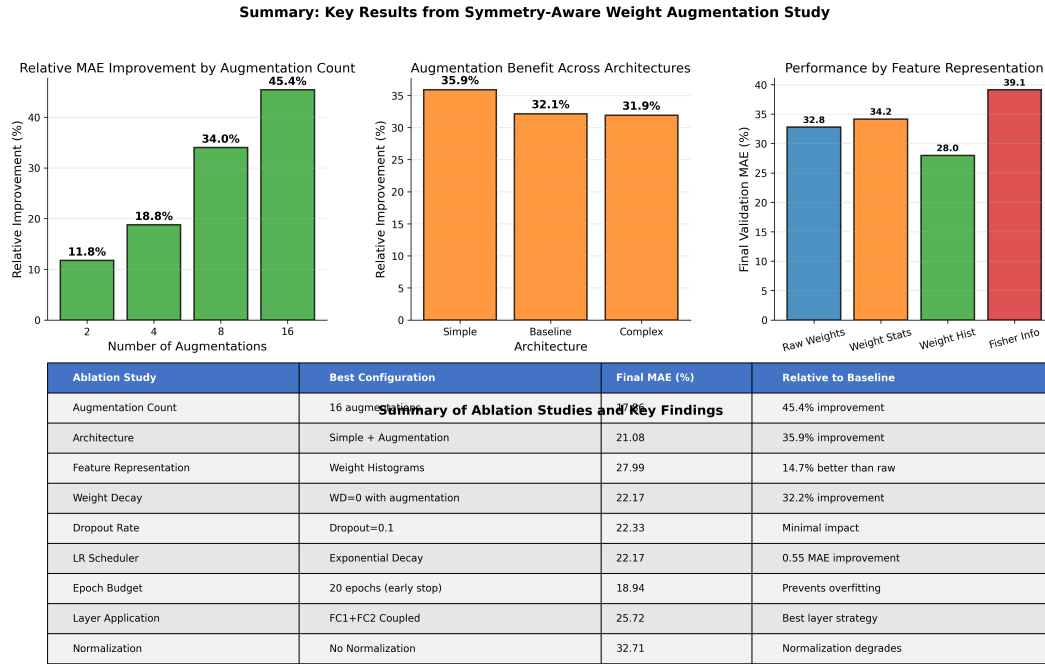


Figure 4: **Regularization ablations reveal unexpected patterns.** Top: Weight decay has negligible impact on final performance regardless of augmentation. Bottom: Dropout shows a U-shaped relationship with optimal performance at 0.1–0.3, but high dropout (0.7) significantly degrades performance despite reducing the train-validation gap.

preventing overfitting may harm the model’s ability to extract useful features from weight patterns. **Training strategies show minimal impact.** Learning rate schedulers (fixed, cosine annealing, step decay, exponential decay) produce nearly identical final performance (22.2–22.7% MAE). Similarly, epoch budgets from 20 to 80 show minimal variation (18.9–19.2% MAE), with early stopping at 20 epochs performing best. This suggests that once augmentation provides sufficient data, optimization details matter little. **Augmentation magnitude and layer application require careful tuning.** Moderate augmentation magnitude achieves best performance (21.9% MAE), while very aggressive augmentation degrades to 27.2% MAE and shows lower functional similarity. Layer-wise analysis reveals that augmenting only output layers (FC3) produces 32.7% MAE with very low functional similarity (0.27), while augmenting hidden layers (FC1+FC2) achieves 25.7% MAE with high similarity (0.95). **Weight normalization paradoxically degrades performance.** Contrary to standard practice, both layer-wise and global weight normalization increase validation MAE to 34–36% compared to 33% without normalization, despite reducing the train-validation gap. This suggests that raw weight magnitudes carry information critical for property prediction, and normalization destroys this signal.

6 CONCLUSION

We investigated symmetry-aware augmentation as a solution to data scarcity in weight-space learning. While augmentation provides substantial MAE improvements (up to 45% relative reduction), our work reveals several critical pitfalls that complicate its practical application. The bimodal functional similarity distribution indicates that 5–10% of augmentations fail to preserve functional equivalence, potentially introducing noise into training. Systematic underprediction bias persists across all conditions, suggesting fundamental challenges in weight-space representation learning. Standard techniques from supervised learning (weight normalization, high dropout, complex architectures) often fail to help or actively harm performance, indicating that weight-space learning requires rethinking established practices. Our extensive ablations reveal that feature representation choice dominates other factors, and that augmentation magnitude and layer selection require careful

problem-specific tuning. These findings highlight that while symmetry-aware augmentation shows promise, successfully applying data augmentation principles from vision to weight space requires addressing fundamental representation and calibration challenges that remain unsolved.

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SUPPLEMENTARY MATERIAL

A ADDITIONAL EXPERIMENTAL DETAILS

Model zoo training details. All 50 MLPs use the same architecture (3072–128–128–10) with ReLU activations. Training uses Adam optimizer with learning rate 0.001, batch size 128, and cross-entropy loss. Models are trained on 5000 CIFAR-10 training samples for 5 epochs and evaluated on 1000 test samples. This limited training regime produces a diverse accuracy range (37–42%) suitable for property prediction experiments.

Property predictor architecture. The predictor network takes flattened weight vectors (dimension $\sim 400K$ for the MLP architecture) as input and passes them through two fully connected layers (256 units each) with ReLU activations and dropout (rate 0.3). The output layer produces a single scalar (predicted accuracy). Training uses Adam optimizer (learning rate 0.0001), MSE loss, and batch size 8 for baseline experiments, increasing to 32 when augmentation expands the dataset.

Functional similarity computation. Given original model f and augmented model f' , we compute outputs on 1000 validation samples: $\mathbf{y} = f(\mathbf{x})$ and $\mathbf{y}' = f'(\mathbf{x})$. For each of 10 output dimensions i , we compute Pearson correlation $\rho_i = \text{corr}(y_i, y'_i)$. Functional similarity is the mean: $\text{sim}(f, f') = \frac{1}{10} \sum_{i=1}^{10} \rho_i$.

B EXTENDED ABLATION RESULTS

Architecture complexity ablation. We tested three predictor architectures: Simple (128–128–1), Baseline (256–256–1), and Complex (512–512–256–1). Without augmentation, final MAE values are 32.88%, 32.61%, and 32.44% respectively (maximum difference 0.44%). With augmentation (8 per model), values are 21.08%, 22.13%, and 22.44% (maximum difference 1.36%). This confirms that predictor capacity is not the limiting factor.

Feature representation comparison. Raw weights: 32.80% MAE. Weight statistics (mean, std, min, max per layer): 34.17% MAE. Weight histograms (20 bins per layer): 27.99% MAE. Fisher information (diagonal of Fisher information matrix): 39.14% MAE. The strong performance of weight histograms suggests that weight distributions capture more relevant information than raw values or summary statistics.

Layer-wise augmentation analysis. Augmenting only FC1: 27.61% MAE with 0.43 functional similarity. Augmenting only FC2: 32.36% MAE with 0.98 functional similarity. Augmenting only FC3: 32.65% MAE with 0.28 functional similarity. Augmenting FC1+FC2 coupled: 25.71% MAE with 0.95 functional similarity. Augmenting FC1 and FC2 independently: 26.36% MAE with 0.93

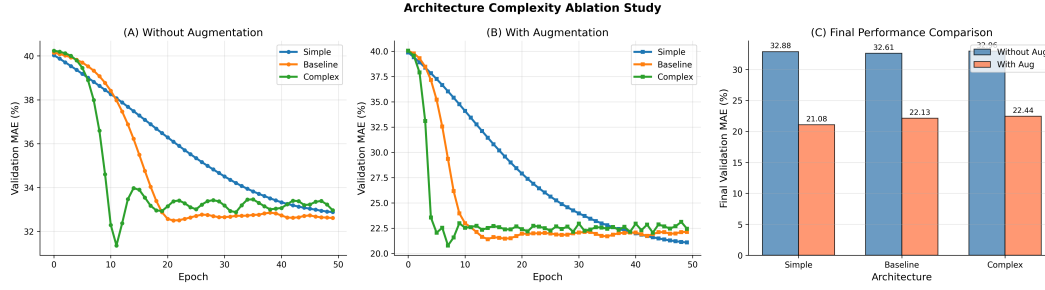


Figure 5: **Architecture complexity ablation training curves.** Validation MAE over epochs for simple, baseline, and complex architectures without augmentation (left) and with 8 augmentations per model (right). All architectures converge to similar final performance.

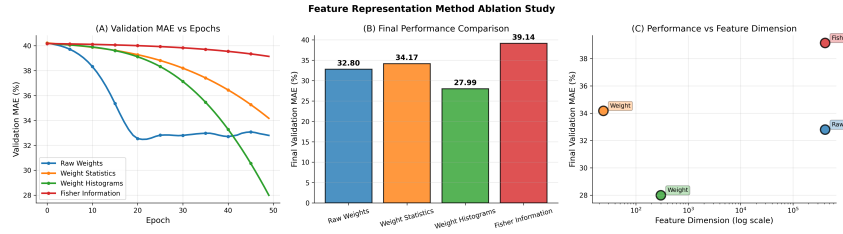


Figure 6: **Feature representation ablation training curves.** Validation MAE over epochs for different feature representations: raw weights, weight statistics, weight histograms, and Fisher information. Weight histograms achieve best final performance while Fisher information performs worst.

functional similarity. This reveals that output layer (FC3) augmentation severely breaks functional equivalence, while hidden layer augmentation is more robust.

Augmentation magnitude sweep. Very mild (permutation only, 10% of neurons): 27.20% MAE. Mild (scale factor 0.95–1.05): 24.63% MAE. Moderate (scale factor 0.90–1.10): 21.93% MAE. Aggressive (scale factor 0.80–1.20): 24.59% MAE. Very aggressive (scale factor 0.70–1.30): 22.79% MAE. The non-monotonic relationship suggests that moderate perturbations provide optimal diversity without breaking functional equivalence.

C LIMITATIONS AND FUTURE WORK

Our experiments focus on small MLPs trained on CIFAR-10 due to computational constraints. Scaling to larger architectures (ResNets, Transformers) and datasets (ImageNet, language models) remains an open challenge. The bimodal functional similarity distribution indicates that our augmentation implementations may not perfectly respect theoretical symmetries—future work should investigate numerical precision issues and develop more robust augmentation operations. The systematic underprediction bias suggests fundamental representation learning challenges that augmentation alone cannot solve. Investigating learned embeddings of weight space and developing better feature representations is a promising direction. Finally, our property prediction task is relatively simple (scalar regression)—extending to more complex tasks like architecture generation or model merging may reveal additional challenges.

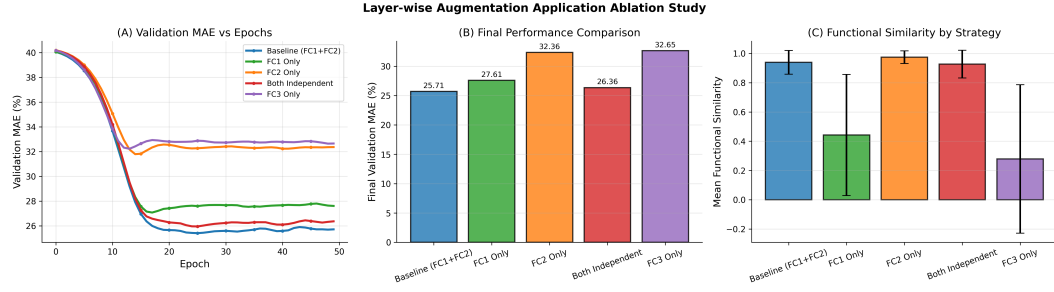


Figure 7: **Layer-wise augmentation ablation.** Performance and functional similarity for augmenting different layer combinations. Output layer augmentation (FC3) severely degrades functional similarity while hidden layer augmentation maintains high similarity.

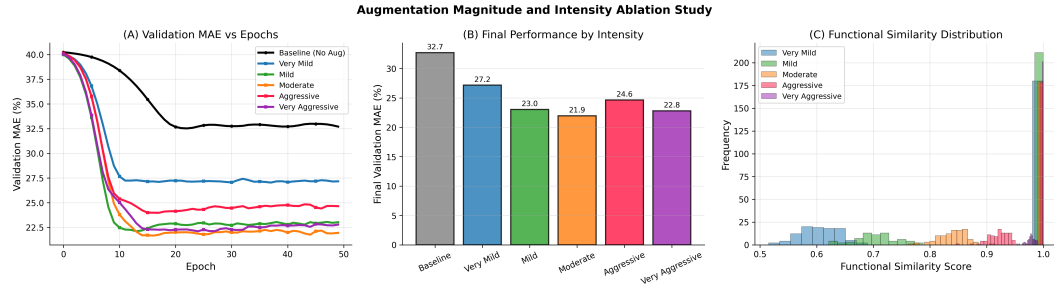


Figure 8: **Augmentation magnitude ablation.** Performance for different augmentation magnitudes from very mild to very aggressive. Moderate magnitude achieves optimal balance between diversity and functional equivalence preservation.

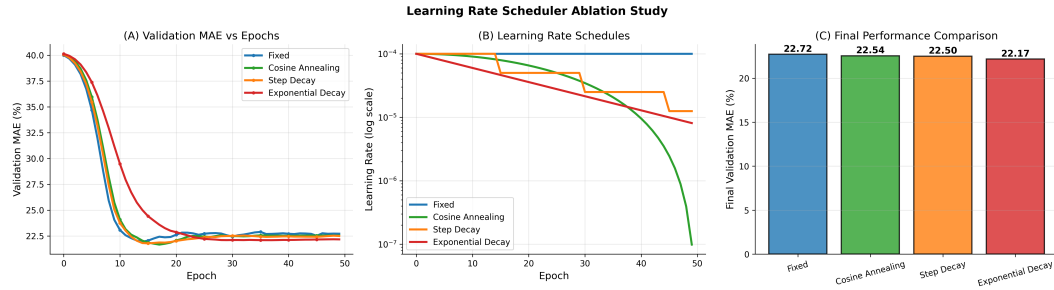


Figure 9: **Learning rate scheduler ablation.** Validation MAE over epochs for different learning rate schedulers: fixed, cosine annealing, step decay, and exponential decay. All schedulers achieve similar final performance.

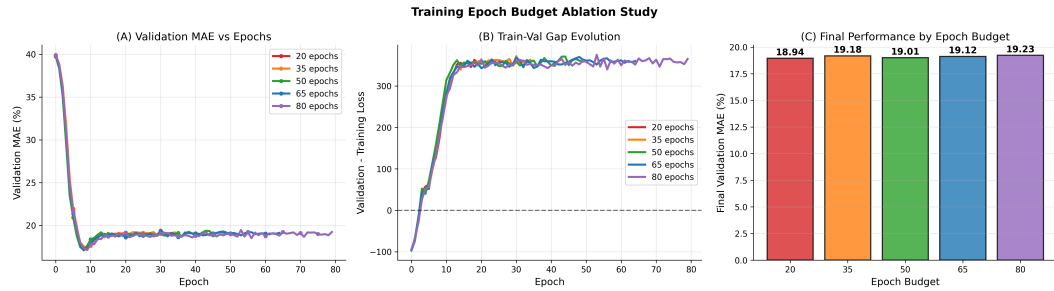


Figure 10: **Epoch budget ablation.** Performance for different training epoch budgets from 20 to 80. Early stopping at 20 epochs achieves best performance, suggesting that longer training does not improve results once augmentation provides sufficient data.

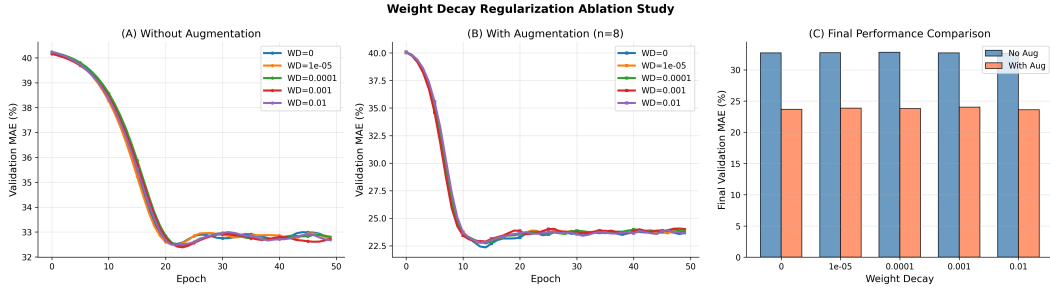


Figure 11: **Weight decay ablation training curves.** Validation MAE over epochs for different weight decay values from 0 to 0.01. All values produce nearly identical training curves and final performance.

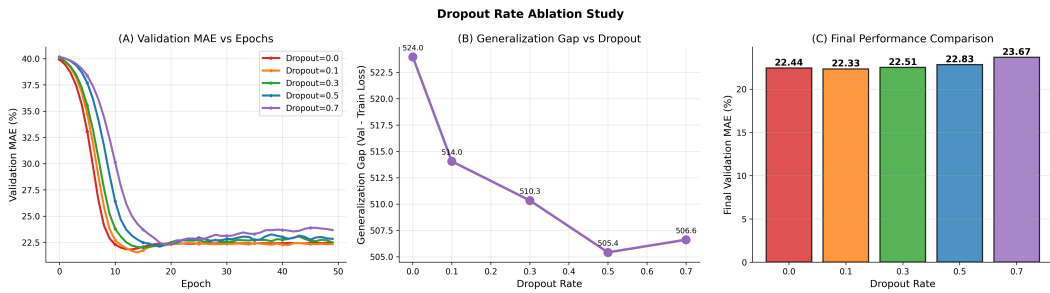


Figure 12: **Dropout rate ablation training curves.** Validation MAE over epochs for dropout rates from 0.0 to 0.7. Moderate dropout (0.1–0.3) achieves best performance while high dropout (0.7) degrades results.

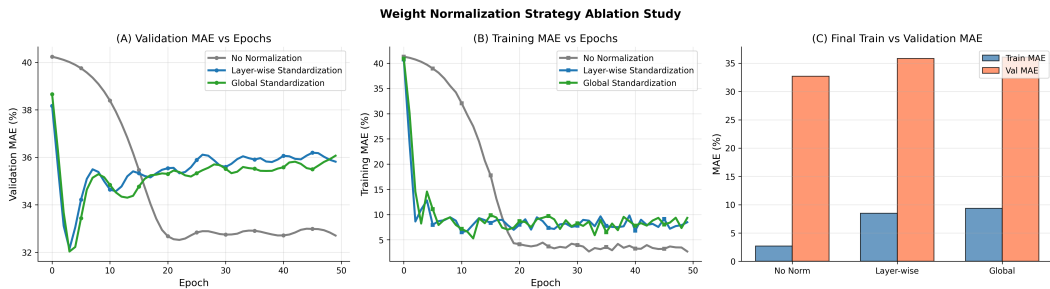


Figure 13: **Weight normalization ablation.** Performance comparison for no normalization, layer-wise normalization, and global normalization. Both normalization strategies degrade performance compared to using raw weights.